

SYNTHGULARITY

BY CALABI YAU

This guide shows you how Synthgularity works, page by page: three wavetable oscillators with double warp, a spectral oscillator, a granular engine, Mutable Instruments modules, eight step sequencers, Lorenz chaos, a scale-aware arpeggiator, two cascade effect racks, a routing page and true-peak and LUFS metering.

VST3 · AU · CLAP · macOS · Windows

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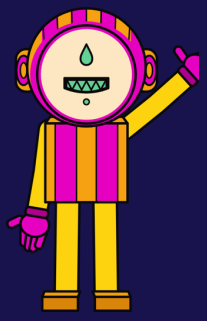
SEQ: eight step lanes

FX: the cascade rack

ROUTING: signal flow, mixer and engine

Three quick recipes

INSTALL



First, let's get me into your DAW. Two minutes, promise.

Unzip the download for your system and copy each plug-in to its folder. Then rescan plug-ins in your DAW (in Ableton Live: Settings > Plug-Ins > Rescan).

FORMAT	FOLDER
macOS AU	/Library/Audio/Plug-Ins/Components/Synthgularity.component
macOS VST3	/Library/Audio/Plug-Ins/VST3/Synthgularity.vst3
macOS CLAP	/Library/Audio/Plug-Ins/CLAP/Synthgularity.clap
Windows VST3	C:\Program Files\Common Files\VST3\Synthgularity.vst3
Windows CLAP	C:\Program Files\Common Files\CLAP\Synthgularity.clap

> On a Mac, files downloaded from the internet are quarantined and the DAW refuses them. Clear the flag once in Terminal:

```
sudo xattr -dr com.apple.quarantine /Library/Audio/Plug-Ins/Components/Synthgularity.compo
```

> Requirements: macOS 10.15 or newer (Apple Silicon and Intel), Windows 10/11 64-bit. Your presets live in Documents/Calabi Yau/Synthgularity/Presets (change it with FOLDERS).

HOW THE SOUND FLOWS

Every voice mixes up to nine sources, bites them, filters them and shapes them. Then the whole synth goes through two effect racks and the meters. The ROUTING page lets you change that path.



WT A · WT B · WT C →

Spectral →

Grain →

Plaits · Sub · Noise · One-shot →

FM / PD / AM matrix →

MultiDistortion →

Filter →

Amp envelope →

FX rack + MIXBUS (10 + 10 slots) →

Master · scope · FFT · dBTP · LUFS

> Modulation (LFOs, Chaos, Mod Envelopes, Seq lanes) reaches almost every knob through the MATRIX page. The arpeggiator sits before the voices and plays them.

✦ HEADER, PRESETS AND MASTER METERS



Up top: presets on the left, and my master section on the right. Watch the numbers while you mix.



ISOLOGO	Click it to open the About card with the Calabi Yau artwork.
PRESETS < > LOAD SAVE INIT	Factory and user presets. LOAD opens any .sgpreset file, from any folder or drive. SAVE stores every knob, the SEQ and ARP steps and the paths of your loaded files. INIT starts from scratch: oscillator 1 (WT A) on and nothing else, no other source, no effects. A new instance opens on INIT too.
FOLDERS	Sets two folders: the one the preset menu lists and LOAD opens (open folder), and the one SAVE writes to (save folder). They can be the same or different, for example a shared library on an external drive. Show this folder opens it in Finder or Explorer; Use the default folders goes back to Documents/Calabi Yau/Synthgularity/Presets. The choice is remembered by every instance on this computer, and if a folder disappears (an unplugged drive) the default is used.
MANUAL	Opens this manual as a PDF, in English or Spanish.
GONIOMETER	Stereo image of the output (vertical = mono, horizontal = side) with a correlation dot: red means phase problems. The full-size version is on the ANALYZER page.
SCOPE	Real-time oscilloscope of the master output, triggered on the waveform, with phosphor trails.
FFT	Live spectrum of the master output, 20 Hz to 20 kHz.
MASTER	Output gain (-36 to +6 dB) before a soft safety limiter that starts at -3 dBFS.

TP (DBTP)	True peak, measured 4x oversampled, so inter-sample peaks show. The bars are L and R; the number is the maximum since the last reset.
M · S · I (LUFS)	Loudness to EBU R128 / ITU BS.1770: Momentary (400 ms), Short-term (3 s) and Integrated (gated, since reset).
RESET	Click the meter numbers to reset TP max and the integrated loudness.
MOD SOURCES (BOTTOM BAR)	Every modulation source as a chip, on every page. Drag one onto any knob (see Drag and drop modulation).

> Quick targets: streaming masters sit near -14 LUFS integrated with true peaks under -1 dBTP. A single synth inside a mix is usually much quieter, and that is fine.

✦ OSC: THREE WAVETABLE OSCILLATORS

A, B and C are identical. Each one reads a table, bends its phase twice with two warp stages, and can modulate the others.



ON · TABLE	Switch the oscillator on and pick a built-in table, or User after LOAD WAV.
POSITION	Scans through the frames of the table.
FREE	On: every note starts at a random phase (wide unison). Off: every note starts at the Phase knob.
WARP 1 · WARP 1 AMT	First warp stage: reshapes the read phase or the sample (see the list below).
WARP 2 · WARP 2 AMT	Second warp stage, applied after the first. Stack them: Bend + Hard Sync, Mirror + Crush...
OCTAVE · SEMI · FINE	Pitch: plus or minus 4 octaves, 12 semitones and 100 cents.

PHASE	Start phase of each note when Free is off.
UNISON · DETUNE · SPREAD	Up to 7 copies, their detune (up to 50 cents) and their stereo spread.
LEVEL · PAN	Output level and position of the oscillator.
LOAD WAV	WAV, AIFF, FLAC or MP3. Files whose length is a multiple of 2048 samples load as frames (Serum style, up to 256); short files become one cycle.

WARP MODES

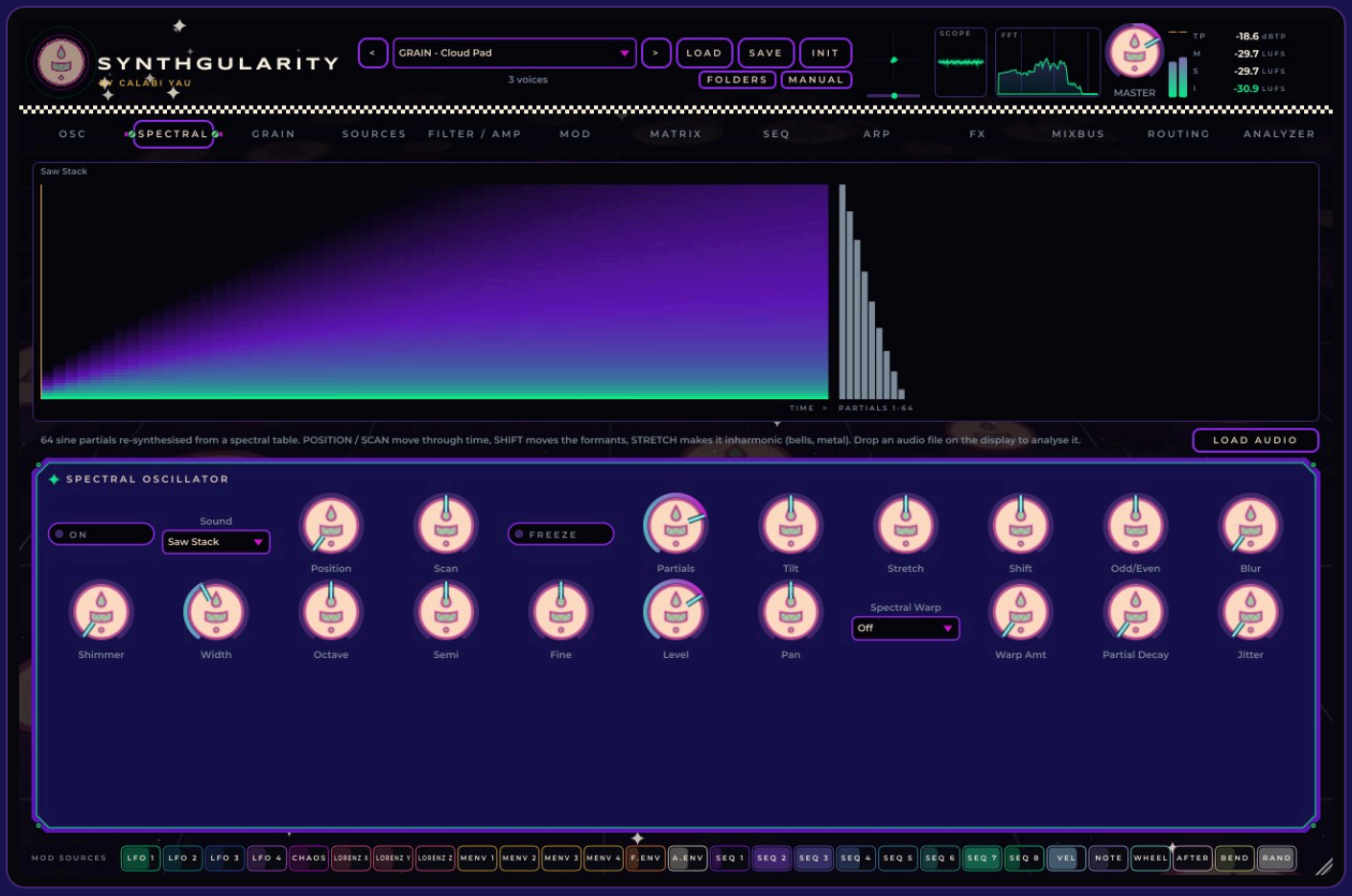
- **Bend + / - / +-** pushes the cycle's energy to one side
- **Bloat + / -** the opposite curve of Bend
- **Hard Sync · Sync Window** re-reads the cycle faster; Window fades each burst
- **Pulse Width** squeezes the first half of the cycle
- **Phase FM · FM Window** self phase-modulation at 2x
- **Flip · Mirror** inverts or mirrors the second half
- **Breed · Brew** adds the octave / a 5x ripple
- **Sinc · Window** sync with a sine or power window
- **Crush · Smooth** bit-crush and phase steps / softer mip level

> The display draws the exact cycle you hear, both warps included.

✦ SPECTRAL: THE SPECTRAL OSCILLATOR



Sixty-four sine partials, re-synthesised from a spectral table. Twenty sounds inside me, or drop any audio file and I will analyse it.



SOUND	20 native sounds or User File (after LOAD AUDIO or dropping a file on the display).
POSITION · SCAN · FREEZE	Where you are in the table (time), how fast each note travels through it (4-second sweep at 1.0, backwards when negative) and a hold.
PARTIALS	How many of the 64 partials sound.
TILT	Darker (left) or brighter (right) spectrum.
STRETCH	Moves partials away from the harmonic series: bells, metal, piano-like stiffness.
SHIFT	Moves the formants (the spectral envelope) up or down 24 semitones without changing pitch.

ODD/EVEN	Removes even partials (right, hollow) or odd ones (left).
BLUR · SHIMMER	Smears the spectrum across neighbours and over time / random partial flicker.
WIDTH	Sends alternating partials left and right.
OCTAVE · SEMI · FINE · LEVEL · PAN	Pitch and output, as on the wavetables.
SPECTRAL WARP · WARP AMT	Shift (adds Hz: inharmonic), Fold (folds the top partials back down), Sieve (keeps every Nth partial), Comb (notches), Spread (detunes pairs), Mirror (flips the spectrum), Squeeze (pulls partials together), Formant Lock (formants stay fixed while you play up and down).
PARTIAL DECAY · JITTER	High partials fade during each note, like a plucked string / a random detune per partial, new on every note.
LOAD AUDIO	Detects the pitch of the file and captures 64 frames across it. Unpitched files keep their spectral colour.

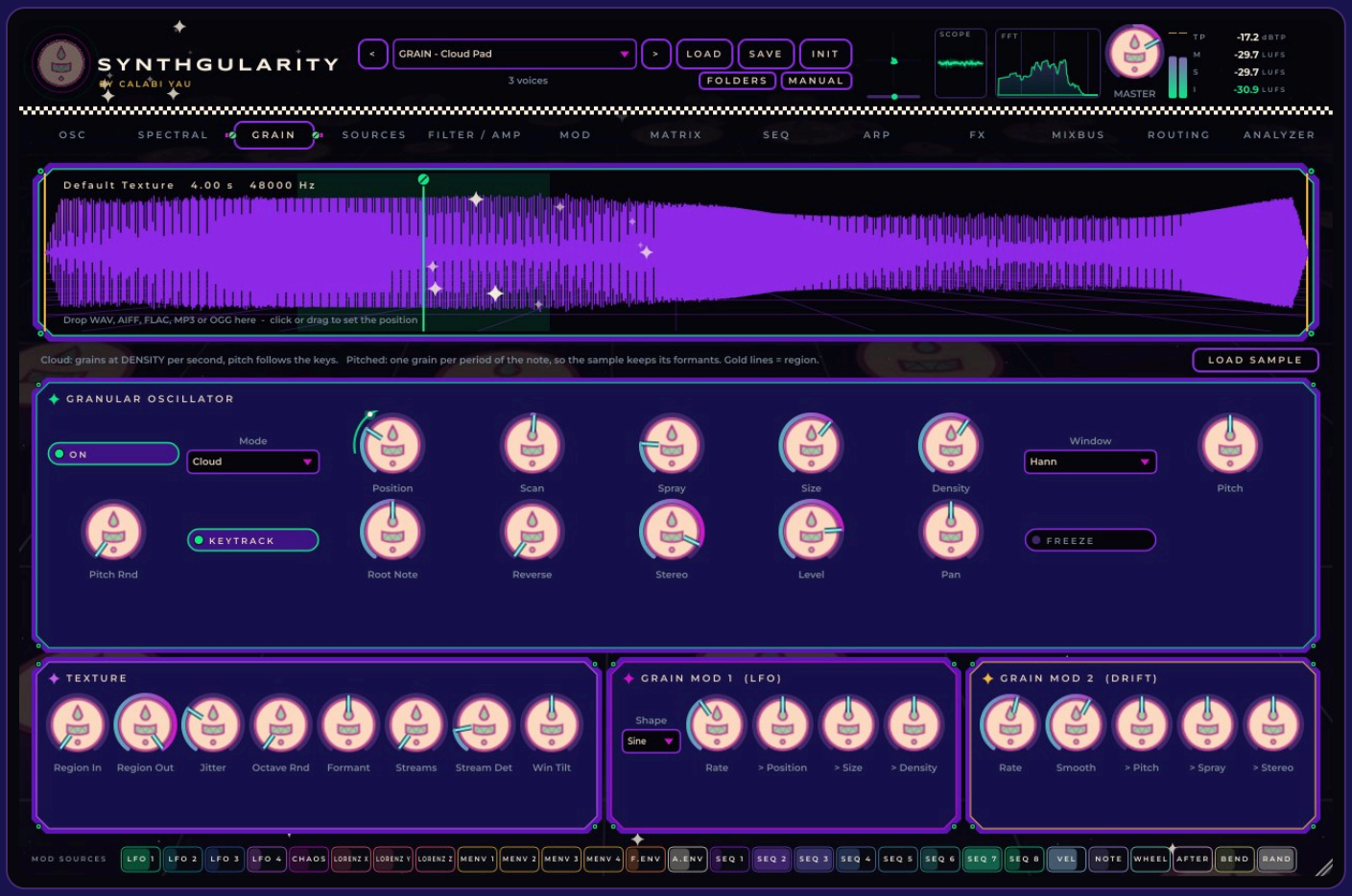
THE 20 SOUNDS

Saw Stack · Hollow Square · Drawbar Organ · Choir Ah-Oh · Vocal Ee-Oo · Bell Bloom · Glass Harmonics · Brass Swell · String Section · Pluck Decay · Bass Growl · Pad Bloom · Formant Sweep · Comb Ripple · Prime Partials · Fibonacci · Odd Shimmer · Spectral Wind · Metal Plate · Singularity

GRAIN: THE GRANULAR OSCILLATOR



Load a WAV, MP3, FLAC, AIFF or OGG and I turn it into clouds of grains. Drag on the waveform to move the position.



MODE

Cloud: grains at DENSITY per second, pitched by the keys. Pitched: one grain per period of the note, so the sample keeps its formants.

POSITION · SCAN · SPRAY

Where grains read, how the read point moves, and random scatter around it.

SIZE · DENSITY · WINDOW

Grain length (5 to 1000 ms), grains per second and their envelope shape.

PITCH · PITCH RND · KEYTRACK · ROOT

Transpose (plus or minus 48 semitones), random pitch per grain, follow the keys from the root note.

REVERSE · STEREO · FREEZE

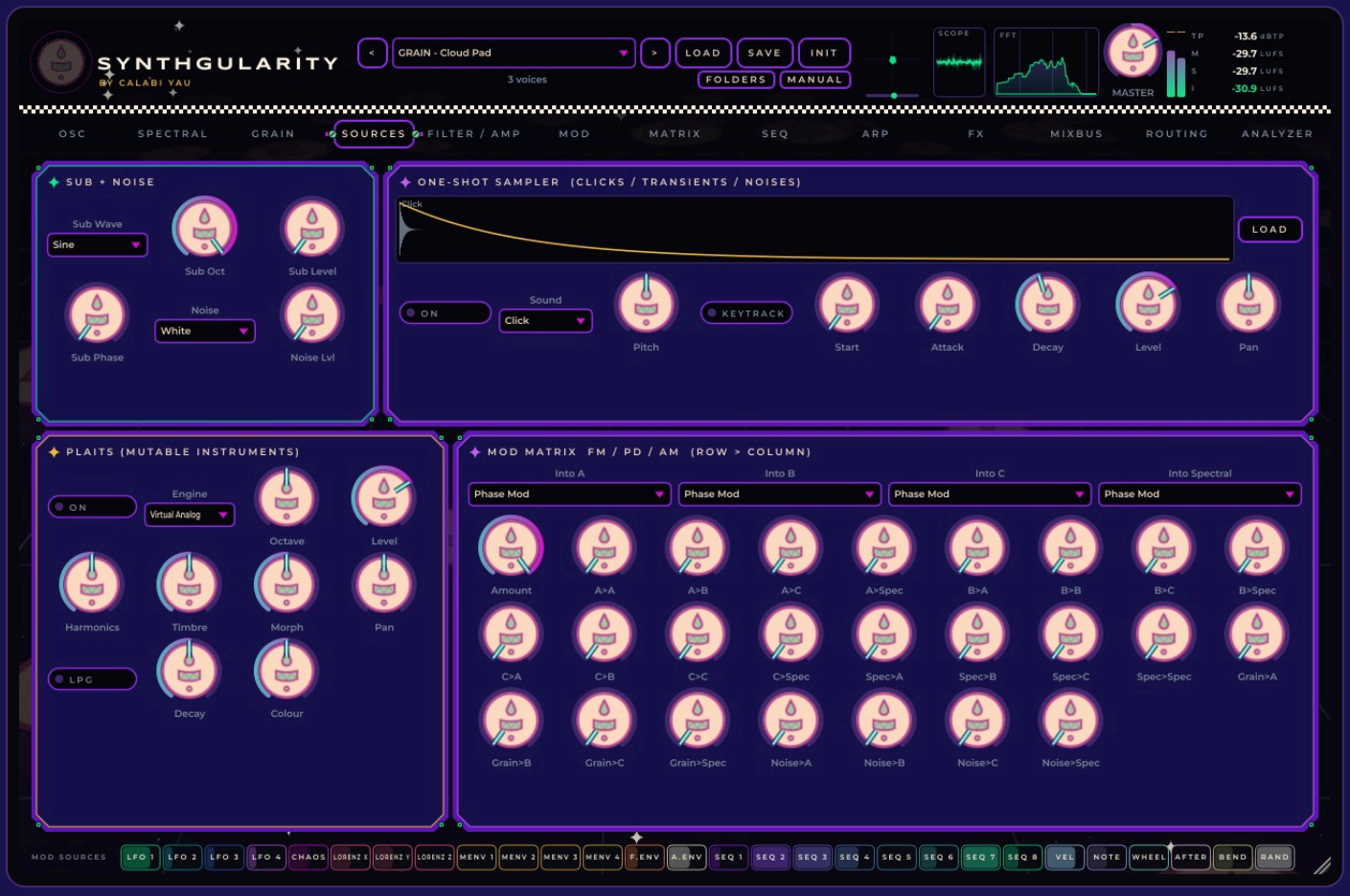
Chance of backwards grains, stereo scatter, and a freeze of the read point.

TEXTURE PANEL	Region in/out, timing jitter, octave jumps, formant, up to 4 detuned streams and window tilt.
GRAIN MOD 1 · 2	A built-in LFO (to position, size, density) and a smooth random drift (to pitch, spray, stereo).

✦ SOURCES: SUB, NOISE, ONE-SHOT SAMPLER, PLAITS AND THE FM MATRIX



Need a click on the attack? The one-shot sampler fires on every note with its own pitch and attack/decay, and skips the amp envelope so it stays sharp.



SUB · NOISE

Sine, triangle or square sub one or two octaves down, with its start phase. White, pink or crackle noise.

ONE-SHOT SAMPLER

12 built-in hits (Click, Tick, Kick Click, Rim, Snap, Noise Burst, Hat, Zap, Blip, Metal Hit, Crackle, Breath) or your own file with LOAD.

PITCH · KEYTRACK · START

Transpose plus or minus 48 semitones, follow the keys from C3, and start later in the sample.

ATTACK · DECAY

The two-stage envelope: linear rise, then a decay to -60 dB. By default the one-shot skips the filter and the amp envelope so clicks stay sharp (change it on ROUTING).

PLAITS	The Mutable Instruments macro-oscillator: 24 engines (VA, FM, chords, speech, drums...), with its own low-pass gate.
FM MATRIX (ROW > COLUMN)	Six sources (A, B, C, Spectral, Grain, Noise) modulate four destinations (A, B, C, Spectral). Type per destination: Phase Mod, Thru-Zero FM, Phase Distortion, Amplitude Mod or Ring Mod. Into Spectral, phase modulation shifts the whole spectrum at once, close to a frequency shifter.

FILTER / AMP

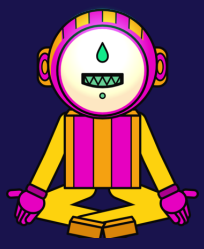


Every voice has its own filter, distortion and five envelopes. Drag the nodes of any envelope to draw it; double-click resets.

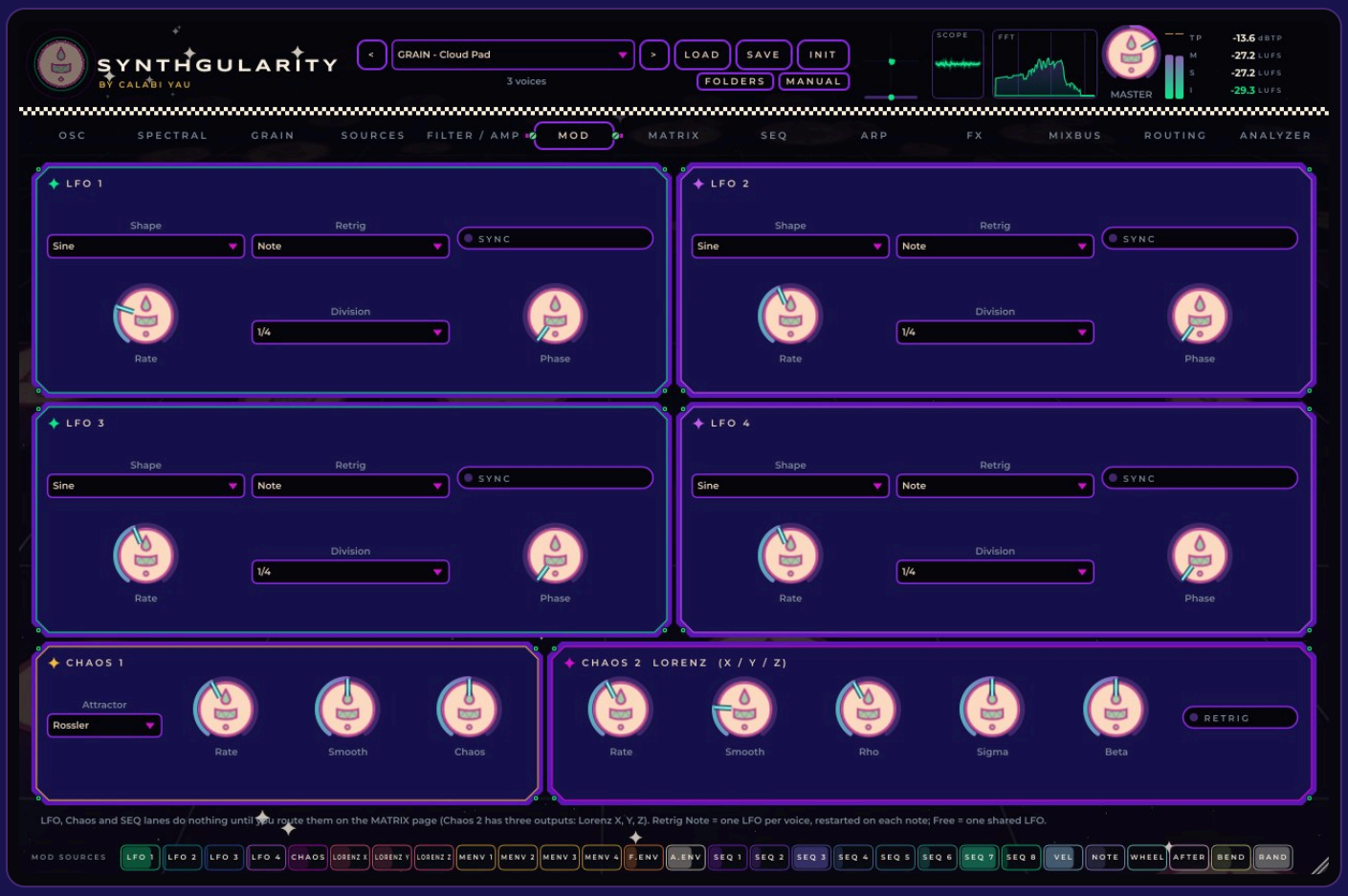


FILTER	Low, band, high pass or notch; Clean SVF or Ladder model; 12 or 24 dB; drive, envelope amount and keytrack.
MULTIDISTORTION	Drive, asymmetry, rectify, wavefold, saturate, bit crush, tone and mix. Post Filter moves it after the filter.
VOICE	Poly, Mono or Legato, number of voices (up to 16), glide and pitch-bend range.
AMP · FILTER ENVELOPE	ADSR for the level and for the cutoff.
MOD ENVELOPE 1-4	Four free ADSRs (tabs 1 to 4). Route them on the MATRIX as Mod Env 1-4.

MOD: LFOS AND CHAOS



My favourite page. Chaos 2 is a Lorenz attractor: three outputs, X, Y and Z, that never repeat.



LFO 1-4

Sine, triangle, saws, square, S&H and smooth random. Retrig Note = one LFO per voice restarted on each note; Free = one shared LFO. Sync locks the rate to the host tempo.

CHAOS 1

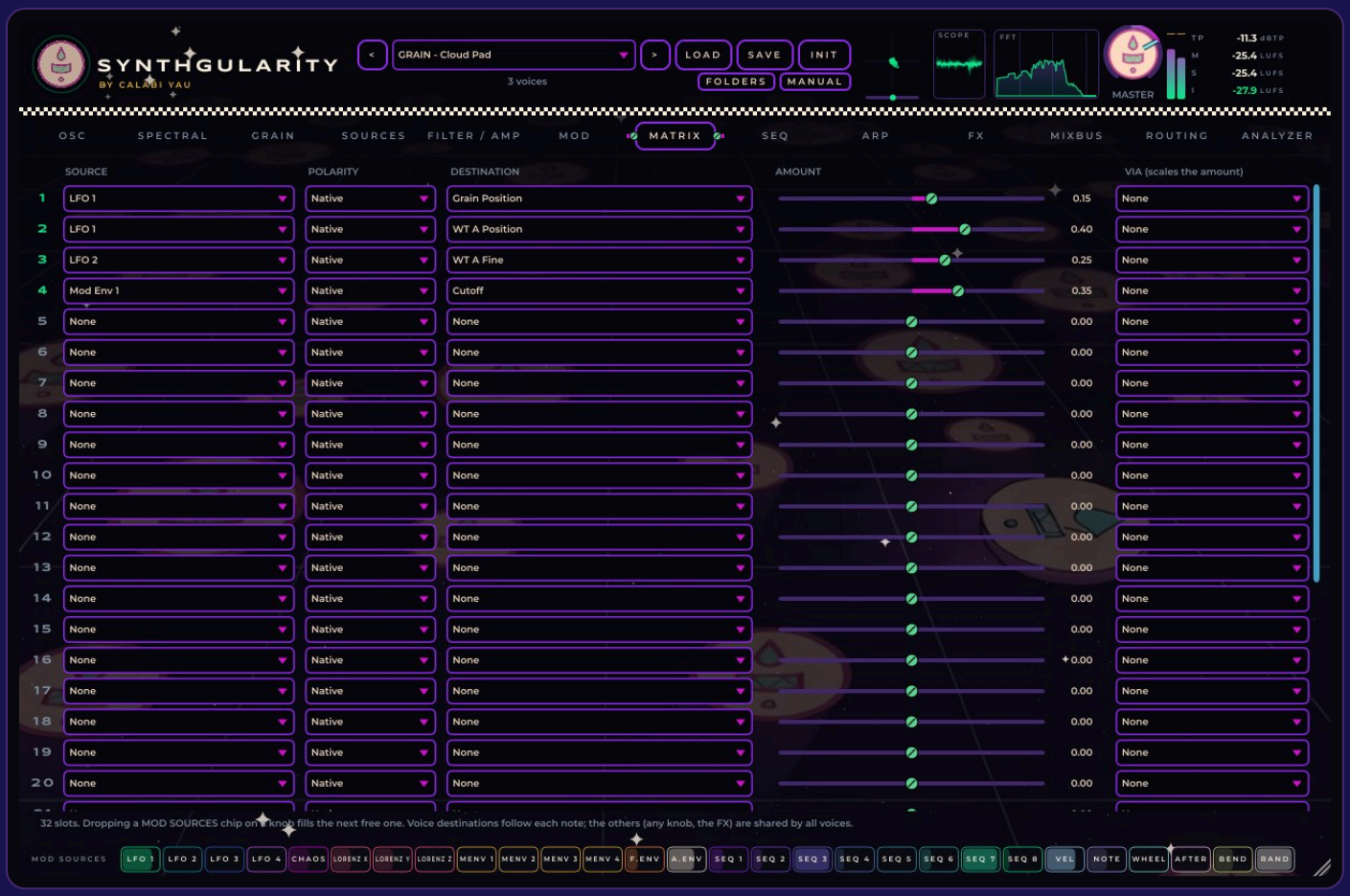
Choose an attractor: Rossler (smooth spirals), Henon and Logistic (stepped maps) or Duffing. Chaos sets how wild it gets.

CHAOS 2 LORENZ

Rate and Smooth, plus the three Lorenz constants: Rho (chaos), Sigma (how fast X follows Y) and Beta (the Z damping). Retrig restarts it on each note.

MATRIX: 32 MODULATION SLOTS

Nothing moves until you route it here. Source, polarity, destination, amount, and a Via source that scales the amount.



SOURCE

LFO 1-4, Chaos 1, Lorenz X/Y/Z, Seq 1-8, Mod Env 1-4, Filter/Amp Env, Velocity, Note, Mod Wheel, Aftertouch, Pitch Bend, Random.

POLARITY

Native keeps the source as it is; Unipolar forces 0 to 1; Bipolar forces -1 to +1.

DESTINATION

Voice / FX destinations: every oscillator (position, pitch up to 4 octaves, level, warp), spectral controls including its warp, grain, sampler, Plaits, distortion, filter, pan, volume, and the FX: delay time/mix/feedback, Rings, Clouds, both filter FX, the reverb and the MIXBUS (DJ filter, gate depth, tape drive, crusher bits, width). Below them, grouped by module: any continuous knob of the synth (WT A Fine, reverb decay, an LFO rate...).

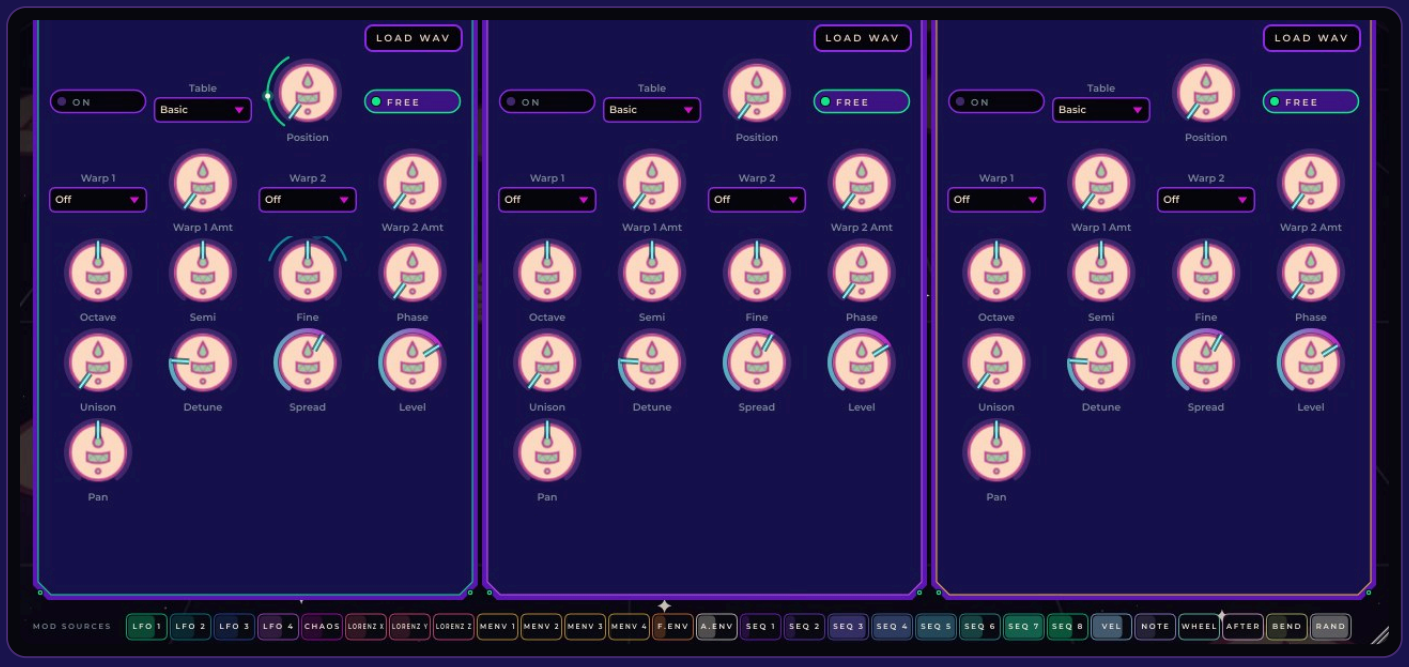
AMOUNT · VIA	Depth from -1 to +1; Via multiplies it by a second source (mod wheel to fade an LFO in, for example).

> Voice destinations follow each note on its own (a filter envelope per note in a chord). The other knobs and the FX are shared by all voices; envelopes, velocity, note and random there follow the newest note.

DRAG AND DROP MODULATION



The fastest way to move a sound: grab a chip from the MOD SOURCES bar and drop it on any knob, on any page.



MOD SOURCES

The bar at the bottom of every page: LFO 1-4, Chaos, Lorenz X/Y/Z, Mod Env 1-4, Filter and Amp Env, Seq 1-8, Velocity, Note, Wheel, Aftertouch, Bend, Random. Each chip shows its own value moving.

DROP

While you drag, every knob that accepts the source lights up. Dropping fills the next free MATRIX slot with 35 % (envelopes, velocity...) or 25 % (LFOs and other bipolar sources).

RINGS

A modulated knob shows one coloured arc per route (the colour of the source) with the range it covers, and a white dot on the value it really has now.

RIGHT-CLICK

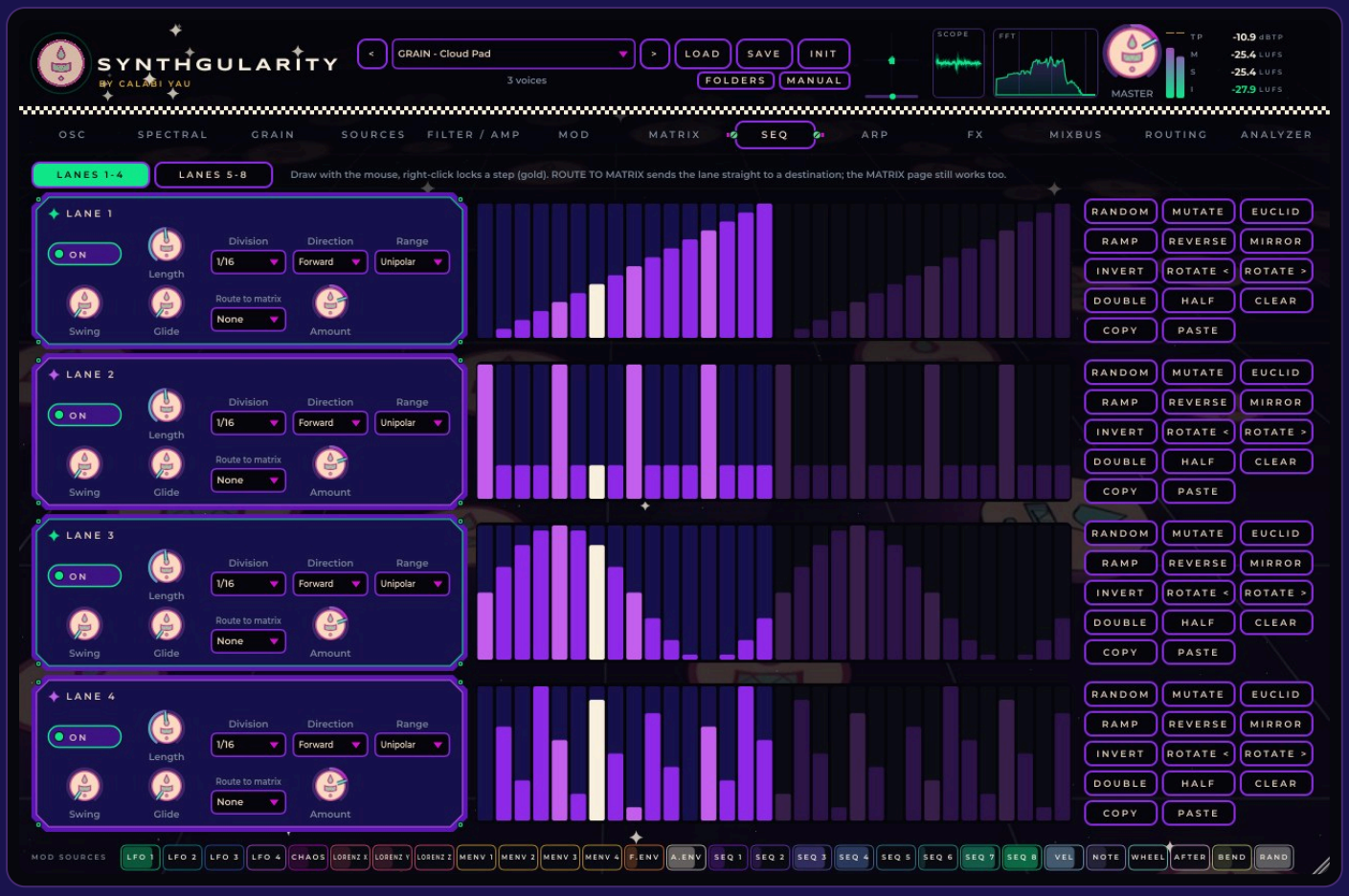
On a knob: amount from -100 % to +100 %, Invert, Remove, or Reset to default. Fine-tune the same routes on the MATRIX page.

Knobs glide to new values in about 12 ms, octave and semitone steps slide, effects fade in and out, and switches that change the voices (a table, an engine, a route) make a 6 ms dip instead of a click. Playing and automating never pops.

SEQ: EIGHT STEP LANES



Eight lanes of up to 32 steps, four on screen at a time. Each lane can run in seven directions.



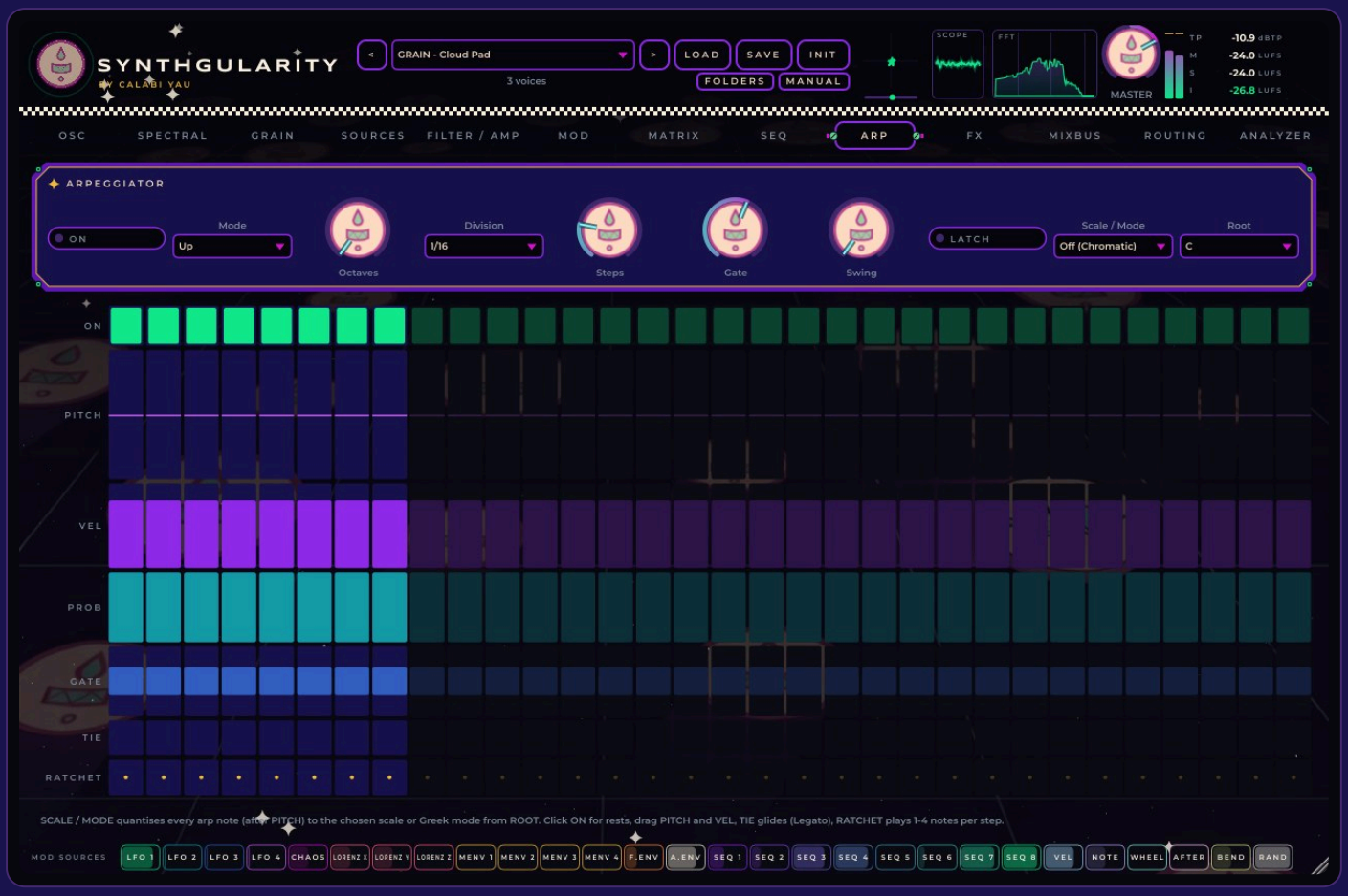
LENGTH · DIVISION · SWING · GLIDE	Steps used, step length synced to the host, swing of the even steps and smoothing between steps.
RANGE	Unipolar (0 to 1) or Bipolar (-1 to +1) output.
DIRECTION	Forward, Backward, Ping-Pong, Random, Even Steps (2-4-6…), Odd Steps (1-3-5…) and Evens > Odds.
ROUTE TO MATRIX · AMOUNT	Send the lane straight to any destination without using the MATRIX page (both work together).
EDITING	Draw with the mouse; right-click locks a step (gold) so RANDOM and MUTATE leave it alone. EUCLID spreads hits evenly; RAMP draws a ramp.

TRANSFORMS	REVERSE, MIRROR (the second half mirrors the first), INVERT (upside down), ROTATE < >, DOUBLE (twice as fast), HALF (twice as slow). COPY and PASTE move a whole lane to another.

✦ ARP: ARPEGGIATOR WITH SCALES AND GREEK MODES



Hold a chord and I play it. With Scale / Mode every note lands inside the scale you pick, even after the PITCH steps.



MODE · OCTAVES · DIVISION

Up, Down, Up/Down, Down/Up, As Played, Random or Chord, over 1 to 4 octaves, synced to the host.

STEPS · GATE · SWING · LATCH

Pattern length, note length, swing and hold after you release the keys.

SCALE / MODE · ROOT

Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Locrian, harmonic and melodic minor, Phrygian dominant, Hungarian minor, double harmonic, pentatonics, blues, whole tone, Hirajoshi and In Sen, from any root.

STEP GRID

ON (rests), PITCH (plus or minus 24), VEL, PROB (chance that the step plays, in 5 % steps), GATE (note length of that step, drawn as a bar to the right), TIE (glide into the next step with Legato) and RATCHET (1 to 4 notes in the step).

GREEK MODES AT A GLANCE

Ionian bright, major

Dorian minor with a raised 6th

Phrygian dark, flat 2nd

Lydian dreamy, raised 4th

Mixolydian major with a flat 7th

Aeolian natural minor

Locrian unstable, flat 2nd and 5th

FX: THE CASCADE RACK

Ten slots, top to bottom. Pick an effect per slot, light its dot to switch it on, and use the arrows to change the order.

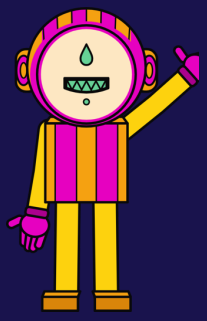


FILTER 1 · FILTER 2	Two filter effects. Freq is modulatable (plus or minus 4 octaves), Spread detunes the right channel.
STEREO	Mono bass below a frequency, M/S width and a Haas delay.
FLANGER · PHASER	Classic modulated effects with feedback.
DELAY	Synced or free time (modulatable from the matrix), ping-pong, feedback filters.
RESONATOR (RINGS) · GRANULAR (CLOUDS)	The Mutable Instruments modal resonator and texture processor.
REVERB	8-line feedback delay network: size, decay up to 20 s, damping, pre-delay, modulation, width and mix.

FILTER FX TYPE	WHAT POLES / MORPH DO
Filterbank	Poles = number of band-pass peaks (1-8); Morph = spacing between them.
Comb +	Positive feedback comb: all harmonics ring. Poles = combs in series; Morph = damping.
Comb -	Negative feedback: only odd harmonics, an octave lower, hollow.
Morph LP>BP>HP	Morph sweeps low pass to band pass to high pass; Poles sets the slope (2 to 8 poles).
Morph LP>Notch>HP	The same through a notch.
Formant (Vowel)	Morph glides A-E-I-O-U; Freq shifts the formants (800 Hz = natural).
Allpass	Poles x 2 allpass stages with feedback (Reso). Mix at 50 % gives phaser notches; Morph spreads the stages.

> The SIGNAL FLOW strip under the editor shows the chain as it runs; effects that are off are greyed. Each effect can sit in one slot only.

MIXBUS: THE SECOND CASCADE



Ten more slots, this time with mix processors for the whole synth. Same rules as the FX rack: menu, dot to switch on, arrows to reorder.



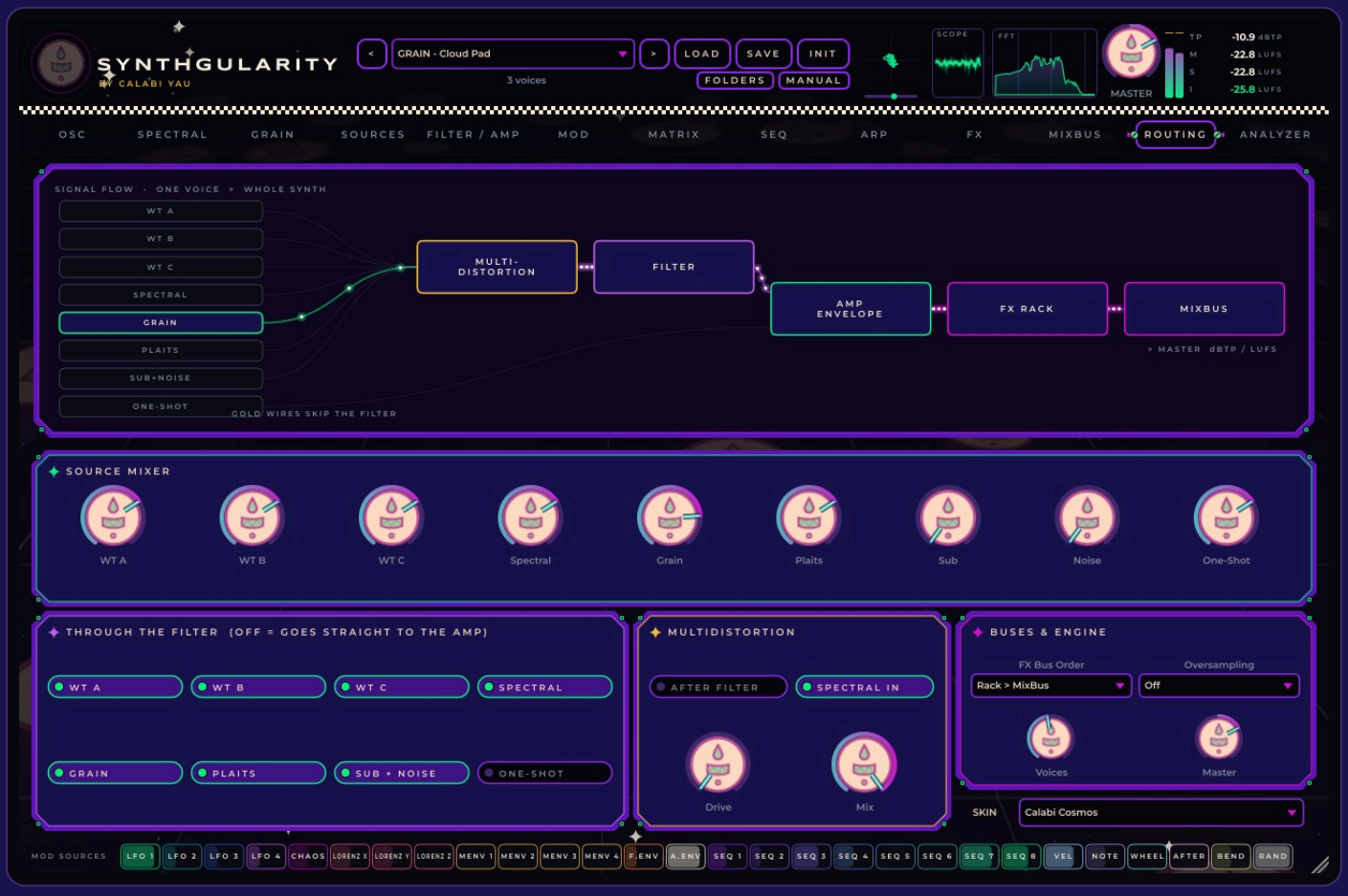
TAPE	Warm asymmetric saturation, tone tilt and wow & flutter.
EQ	Low cut, low shelf, a sweepable mid band, high shelf and high cut.
COMPRESSOR	Stereo-linked with a soft knee: threshold, ratio, attack, release, makeup and mix (parallel compression).
CHORUS · IMAGER	Three-voice stereo chorus / stereo width with mono bass below a frequency.
TRANCE GATE	Eight 16-step patterns locked to the tempo (Straight, Offbeat, Psy Roll, 3+3+2…), with depth and smoothing.
DJ FILTER	One knob: left is low pass, right is high pass, centre is off.

BITCRUSHER · TRANSIENT · LIMITER	Bit depth and downsampling / attack and sustain shaping / a peak limiter with input gain and ceiling.

ROUTING: SIGNAL FLOW, MIXER AND ENGINE



Here you decide the path. The diagram on top updates as you switch things, and the dots show where the sound is flowing.



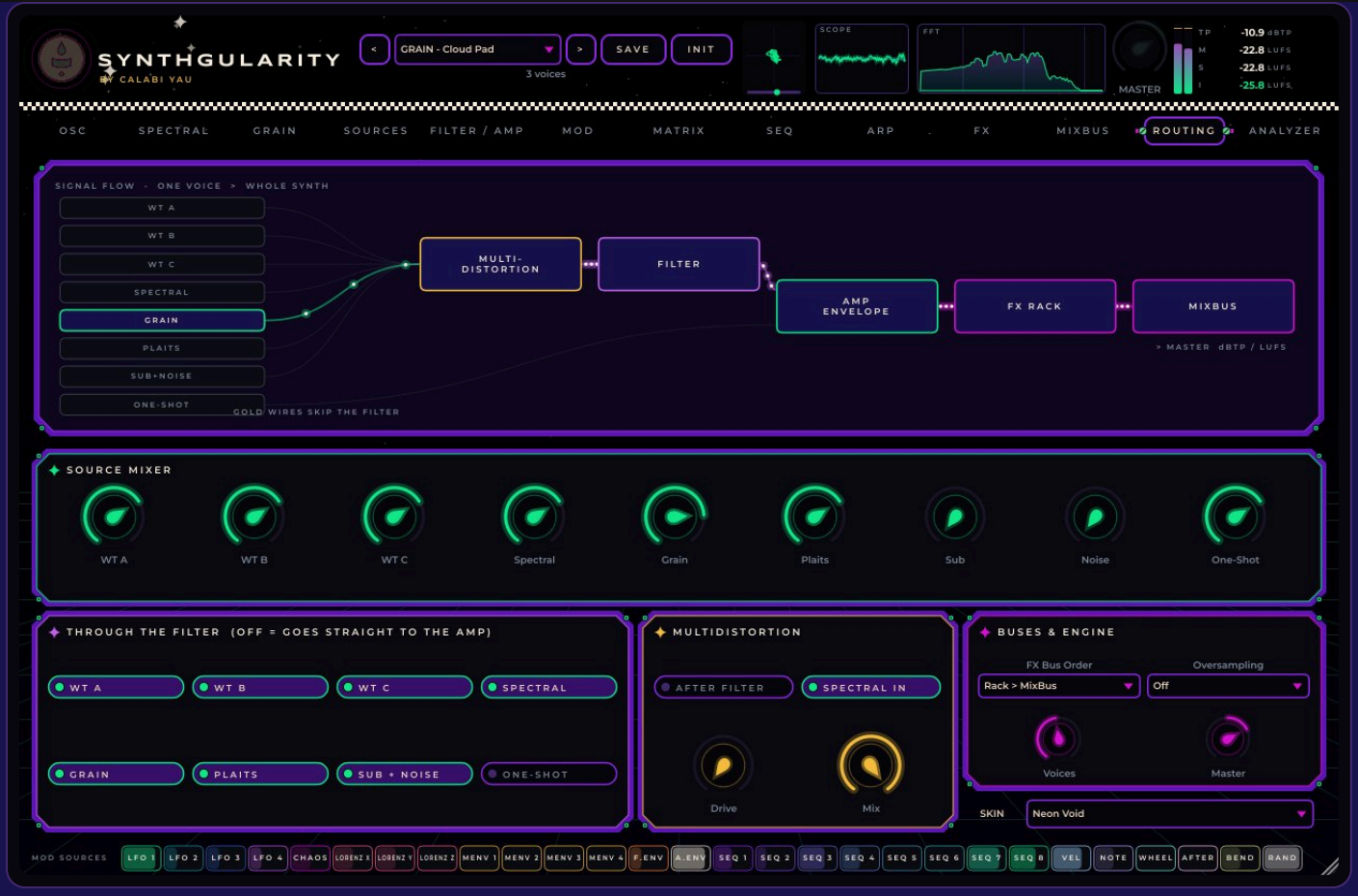
SOURCE MIXER	The level of all nine sources in one row.
THROUGH THE FILTER	Per source: on goes through distortion and filter; off goes straight to the amp (gold wires in the diagram). Great for keeping a sub or a click untouched.
MULTIDISTORTION	Before or after the filter, and whether the Spectral oscillator goes through it.
FX BUS ORDER	Rack > MixBus or MixBus > Rack.
OVERSAMPLING	Off, 4x or 8x: the voices run at a higher rate and are filtered down, so hard warps, FM and distortion alias less. It costs CPU, roughly 4 or 8 times.

SKIN

Calabi Cosmos, Analog Studio (black knobs with aluminium caps) or Neon Void (drop-shaped knobs). Saved with your session.

> The two other skins, Analog Studio and Neon Void, are shown in the pictures below.




SYNTHGULARITY
BY CALABI YAU
GRAIN - Cloud Pad
3 voices
SAVE
INIT
SCOPE
FFT
MASTER
TP -10.9 dBTP
M -22.8 LUFS
S -22.8 LUFS
I -25.8 LUFS
OSC SPECTRAL GRAIN SOURCES FILTER / AMP MOD MATRIX SEQ ARP FX MIXBUS ROUTING ANALYZER
SIGNAL FLOW · ONE VOICE > WHOLE SYNTH
WT A
WT B
WT C
SPECTRAL
GRAIN
PLAITS
SUB+NOISE
ONE-SHOT
GOLD WIRES SKIP THE FILTER
MULTI-DISTORTION
FILTER
AMP ENVELOPE
FX RACK
MIXBUS
> MASTER dBTP / LUFS
SOURCE MIXER
WT A
WT B
WT C
Spectral
Grain
Plaits
Sub
Noise
One-Shot
THROUGH THE FILTER (OFF = GOES STRAIGHT TO THE AMP)
WT A
WT B
WT C
SPECTRAL
GRAIN
PLAITS
SUB + NOISE
ONE-SHOT
MULTIDISTORTION
AFTER FILTER
SPECTRAL IN
Drive
Mix
BUSES & ENGINE
FX Bus Order
Rack > MixBus
Oversampling
Off
Voices
Master
SKIN
Neon Void
MOD SOURCES
LFO 1 LFO 2 LFO 3 LFO 4 CHAOS LORENZ X LORENZ Y LORENZ Z MENV 1 MENV 2 MENV 3 MENV 4 F.ENV A.ENV SEQ 1 SEQ 2 SEQ 3 SEQ 4 SEQ 5 SEQ 6 SEQ 7 SEQ 8 VEL NOTE WHEEL AFTER BEND RAND

ANALYZER



Four big displays of the output, after the effects and the master.



SCOPE	Oscilloscope locked to the rising zero crossing, with phosphor trails.
SPECTRUM	FFT from 20 Hz to 20 kHz, with a slow gold line holding the peaks.
GONIOMETER	L/R rotated 45 degrees: a vertical line is mono, a wide cloud is a wide stereo image. The bar below is the correlation; red means parts of the sound cancel in mono.
OUTPUT	True peak bars (gold line at -1 dBTP) and loudness M, S and INT in LUFS. Click to reset.

✦ THREE QUICK RECIPES



Try these and then break them. That is what I am for.

PSY CLICK BASS

WT A Basic one octave down, Warp 1 Bend + at 40 %, Warp 2 Hard Sync at 30 %. One-Shot Kick Click with 80 ms decay. Voice Mono, Ladder filter at 900 Hz with a short filter envelope. Matrix: Mod Env 1 > WT A Warp.

LORENZ DORIAN ARP

ARP on, Random, 3 octaves, 1/16, Scale Dorian from A. Matrix: Lorenz X > Cutoff (0.3), Lorenz Z > WT B Position (0.4). Delay 1/8T and the reverb in the rack.

SPECTRAL CHOIR

SPECTRAL on, Choir Ah-Oh, Scan 0.15, Blur 0.5, Width 0.8. Filter 1 in Formant mode at 40 % mix. Matrix: LFO 1 > Spectral Shift (0.2). Reverb mix 35 %.

> These three are also factory presets: BASS - Click Growl, ARP - Dorian Lorenz and SPECTRAL - Choir Bloom.

Synthgularity v0.6.1 by Calabi Yau. Plaits, Rings and Clouds DSP by Emilie Gillet (Mutable Instruments, MIT licence). Built with JUCE and the CLAP extensions. Fonts: Syncopate (Apache 2.0) and Montserrat (OFL).